What is the difference between relational algebra and SQL?

Ans:

Point	Relational Algebra	SQL
Type	Procedural	Non-procedural
What it tells	**How** to retrieve data	**What** data to retrieve
Purpose	Theoretical language	Practical language
Use	Foundation for database queries	Used to query real databases
Syntax	Mathematical notation	English-like statements

Write names of 2 aggregate functions with their syntax in relational algebra.

Ans:

Two aggregate functions in relational algebra:

1. **COUNT**
   Syntax: γ count(attribute)(Relation)
2. **SUM**
   Syntax: γ sum(attribute)(Relation)

These functions are used with the **gamma (γ)** operator for aggregation.

Write down the problems of using Cartesian Product. How could it be resolved? Explain with example.

Ans:

Problems of using Cartesian Product:

1. **Large Output:** It combines every row of one table with every row of another, creating a huge number of results.
2. **Irrelevant Data:** Most combinations are meaningless without a condition (like a join).
3. **Performance Issues:** More time and memory needed due to large result size.

Solution: Use a Join instead of Cartesian Product.

Example:

Let's say we have:

- **Student(SID, Name)**
- **Enroll(SID, Course)**

Using Cartesian Product:
Student × Enroll — gives all combinations, many unrelated.

Better way (using Join):
Student ⋈ Student.SID = Enroll.SID Enroll — only meaningful combinations are returned (matching SID).

Why should we avoid natural join? Explain with example

Ans:

Why avoid Natural Join:

1. **Unexpected Results:** It joins tables using all columns with the same name, which might include unwanted columns.
2. **Column Name Conflicts:** If two tables have same-named columns with different meanings, it gives wrong results.
3. **Hard to Debug:** It's not clear which columns are used for joining.

Example:

Tables:

- **Employee(EID, Name, Dept)**
- **Department(Dept, Location)**

Natural Join:
Employee ⋈ Department — joins on **Dept**, which is okay here.

But if both tables also had a **Name** column:

- **Employee(Name, Dept)**
- **Department(Name, Location)**

Then Employee ⋈ Department would join on both **Name** and **Dept**, possibly giving **wrong or no results**.

What is Relational Algebra? Write the difference between inner join and outer join.

Ans:

Relational Algebra:
It is a **formal language** used to **query and manipulate data** in a relational database using mathematical operations.

Difference between Inner Join and Outer Join:

Feature	**Inner Join**	**Outer Join**
Matching Rows	Returns **only matching** rows	Returns **matching + non-matching** rows

Missing Data	**Ignores** unmatched rows	**Includes** unmatched rows with NULLs
Types	Only one type	Left, Right, and Full Outer Joins

Example:

- **Inner Join:** Students who have enrolled in a course
- **Outer Join:** All students, including those **not enrolled** (with NULL for course info)

State the names of basic operators of Relational Algebra with symbols and meaning.

Ans:

Basic Operators of Relational Algebra:

1. **Selection (σ)** – Filters rows based on condition
   Example: σ_{age > 18}(Student)
2. **Projection (π)** – Selects specific columns
   Example: π_{Name, Age}(Student)
3. **Union (∪)** – Combines rows from two relations
   Example: A ∪ B
4. **Set Difference (−)** – Rows in one relation but not in other
   Example: A − B
5. **Cartesian Product (×)** – Combines all rows of two relations
   Example: A × B
6. **Rename (ρ)** – Renames relation or attributes
   Example: ρ_{NewName}(Student)
7. **Join (⋈)** – Combines related rows from two relations
   Example: A ⋈_{A.id = B.id} B
